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UniMo: Unifying Human and Animal Motion Generation

Parametric Skeletons Block Cross-Species Unification — Topology-Agnostic Point Clouds and a Dataset 102× Larger Than Any Prior Animal Corpus Enable a Single Model for All Species

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Text-driven 3D motion generation has matured rapidly for humans, with standardized SMPL skeletons enabling a community of comparable models. Animals remain largely excluded: a dog has 17 joints, a horse 30, a bird 15 — their skeletal topologies are incompatible with each other and with SMPL, forcing per-species model training. UniMo eliminates this barrier by converting any skeleton to a fixed-size unparametric point cloud, training a single diffusion model across species, and releasing UniML3D, a dataset that is 102 times larger than any existing animal motion corpus.

AT A GLANCE

Problem: Diverse animal skeletal topologies prevent training a single motion generation model across species.

Why it matters: Per-species models are expensive; animal motion datasets are tiny (<1,500 sequences); SIGGRAPH applications demand diverse creatures.

Method: Convert parametric skeletons to $N = 256$ point clouds with dynamic sampling; train single diffusion model; IK reconstruction per species at test time.

Dataset: UniML3D: 145,907 sequences + 433,388 captions across human and animal categories.

Result: State of the art on HumanML3D, KIT-ML, and AnimalML3D simultaneously in one model.

Why Existing Approaches Fall Short

Motion generation models represent poses as joint rotation/position vectors in species-specific fixed order:

$$\mathbf{P}_{\text{human}} \in \mathbb{R}^{22 \times 3}, \quad \mathbf{P}_{\text{dog}} \in \mathbb{R}^{17 \times 3}, \quad \mathbf{P}_{\text{horse}} \in \mathbb{R}^{30 \times 3}$$

Dimension and joint semantics differ across species; concatenating these into a single input space is undefined

without topology alignment that does not exist. The result: every new species requires a separate model trained from scratch.

Core Mechanism: Point Cloud Representation

UniMo samples $N = 256$ points from the bones of any skeleton, regardless of species topology. For each bone b connecting joints j_a and j_b , n_b equally-spaced points are placed along the bone segment. The pose at time t is:

$$\mathbf{P}_t \in \mathbb{R}^{N \times 3}, \quad N = \sum_b n_b$$

This representation is topology-agnostic (always 256 points), continuous (captures limb geometry, not just endpoints), and differentiable (linear interpolation along bones).

Dynamic sampling. Bone activity at time t :

$$v_{b,t} = \|\Delta \mathbf{p}_{j_a,t}\|_2 + \|\Delta \mathbf{p}_{j_b,t}\|_2$$

Mean activity: $\bar{v}_b = \frac{1}{T} \sum_t v_{b,t}$. Point allocation:

$$n_b = \max\left(1, \left\lfloor N \cdot \frac{\bar{v}_b}{\sum_{b'} \bar{v}_{b'}} \right\rfloor\right)$$

This concentrates the N -point budget on actively moving bones, sharpening the representation of the most informative parts of motion.

Technical Formulation

Diffusion forward process:

$$\mathbf{P}_t^{(\tau)} = \sqrt{\alpha_\tau} \mathbf{P}_t^{(0)} + \sqrt{1 - \alpha_\tau} \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, I)$$

Denoising objective:

$$\mathcal{L} = \mathbb{E}_{\tau, \boldsymbol{\epsilon}} \left[\left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{P}^{(\tau)}, \tau, \mathbf{c}) \right\|^2 \right]$$

where $\mathbf{c}$ is a CLIP text embedding and $\boldsymbol{\epsilon}_\theta$ is a Transformer denoiser with cross-attention to $\mathbf{c}$.

Inverse kinematics reconstruction: Given predicted point cloud $\hat{\mathbf{P}}_t$, recover joint rotations by minimizing:

$$\hat{\boldsymbol{\theta}}_t = \arg \min_{\boldsymbol{\theta}} \sum_b \left\| \text{FK}(\boldsymbol{\theta})_b - \hat{\mathbf{P}}_{t,b} \right\|^2 + \lambda \sum_b (\|\Delta \text{FK}_b\|_2 - \ell_b)^2$$

where FK is forward kinematics and ℓ_b is rest-pose bone length.

UniML3D Dataset

Category	Sequences	Captions
Human	128,319	384,957
Animal (quadruped)	14,231	42,693
Animal (avian)	3,357	5,738
Total	145,907	433,388

The largest prior animal dataset contains $\sim 1,432$ sequences; UniML3D is $102\times$ larger.

Experimental Results

Method	HumanML3D FID↓	KIT-ML FID↓	AnimalML3D FID↓
MDM	0.544	0.497	—
MotionDiffuse	0.630	1.954	—
MLD	0.473	0.404	—
Species-specific	—	—	2.14
UniMo	0.412	0.381	1.07

UniMo improves over MLD on both human benchmarks while being the first unified model to achieve competitive quality on animals.

Ablations and Interpretation

Without dynamic sampling (static uniform): AnimalML3D FID degrades by 0.11; most degradation on fast-moving quadruped sequences.

Without joint training on animals: AnimalML3D FID = 4.32 (near random), confirming human-only training does not generalize.

Point count N : Plateaus at $N = 256$; $N = 128$ degrades FID by 0.09 on animals; $N = 512$ shows no improvement.

IK reconstruction: Mean bone-length error of 1.2 mm ($< 0.1\%$ of body height), confirming the IK solver accurately recovers valid joint configurations from predicted point clouds.

Reference

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